

Tyler Burguillos

Austin Nolte

Alan Lei

Nishan Majarjan

October 6, 2024

Professor Haiquan Chen

CSC 180 - Intelligent Systems

Due Date: October 9, 2024

Project 2 Report

Problem Statement:

For this project, the goal is to detect network intrusions using the UNSW-NB15 dataset. This involves building both Fully-Connected Neural Networks (FNN) and Convolutional Neural Networks (CNN) to classify network traffic as either normal or malicious. We'll train our models, tune hyperparameters like optimizers, learning rates, and layer configurations, and evaluate their performance using accuracy, precision, recall, and F1-score. As an additional feature, .

Methodology:

We used the keras tuner to find the hyperparameters of the models. We then ran a model for each different activation function for the last layer. We then fine tuned the best performing model to the best of our ability.

Experimental Results and Analysis:

Through all of our struggles we found that for some reason, all of our models were giving the same output in the confusion matrix. It was only biased towards True Negatives and False

negatives. This was for reasons unbeknownst to us and through all of our struggles we could not get the models to do anything but give the exact same results. We tried changing the shape of the data, dropping different columns, encoding different columns. No matter what we tried it seemed to always give the exact same results. We always ended up with an F1 score of .43 for normal with a precision of .27 and a recall of 1.00. We would end up with a 0 across the board for all three of those in the attack portion of the predictions.

Task Division and Project Reflection:

We originally created and agreed upon a document that would generally split up the work while also allowing for adjustments depending on each person. One of the issues we faced was that when we discovered that we were using 2 neurons in the output layer rather than one, we changed the code in response in order to compensate. However, this caused many other errors to occur.